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Instruction Granularity as Planning Width: Formalizing and Benchmarking for Vision-Language-Action Models

Author Name

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Reviewed as if submitted to IJCAI 2023 - main track ⓘ

Disclaimer: Only the main paper is reviewed; anonymity and formatting are not checked. The desk-rejection assessment evaluates content completeness, correctness, novelty, clarity, and scientific quality in line with common IJCAI review guidelines.

Desk Rejection Assessment:

Paper Length

Pass

Topic Compatibility

Pass . The paper focuses on Vision-Language-Action (VLA) models, instruction granularity, and planning width, which aligns with IJCAI topics such as "Machine Learning," "Intelligent Robots," "Natural Language Processing," and "Planning, Routing & Scheduling."

Minimum Quality

Pass . The paper contains all required sections (Abstract, Introduction, Related Work, Methodology, Experiments, Conclusion). It presents a novel formalization,

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Expected Review Outcome:

Summary

This paper addresses the impact of instruction granularity on Vision-Language-Action (VLA) models by formalizing granularity as *planning width* (specifically rule width), adapted from classical width-based planning. To systematically evaluate this, the authors introduce the *Mini-BEHAVIOR-Gran* benchmark, which augments complex household tasks with instruction sets at varying widths and a dynamic instructor. The study reveals a counter-intuitive U-shaped performance curve where models perform best with either highly fine-grained or highly coarse instructions, while struggling with medium granularity. Additionally, the paper uncovers an asymmetry in generalization—coarse-trained models generalize to fine instructions better than the reverse—and provides a diagnosis of "language ignoring" behaviors linked to instruction predictability.

Detailed Feedback

Strengths

- **Novel Theoretical Formalization:** The adaptation of *rule width* from classical planning to define "instruction granularity" is a conceptually strong and rigorous contribution. By defining granularity based on the number of simultaneous features an agent must track (**Equation 1, Page 3**), the paper moves beyond vague definitions like "level of detail" or "sentence length." This provides a quantifiable metric that is grounded in the complexity of the underlying search problem.
- **Insightful Empirical Findings:** The discovery of the U-shaped performance curve (**Figure 3**) is significant and challenges the linear intuition that "more detail is always better" or "coarser is harder." The analysis successfully disentangles instruction length from planning width (**Figure 5** and **Figure 6**), providing strong evidence that the difficulty stems from the planning demand rather than just sequence length.
- **Analysis of Generalization and Language Ignoring:** The finding that training on coarse instructions promotes better cross-granularity generalization (**Figure 4**) is practically valuable for VLA curriculum design. Furthermore, the analysis of "language ignoring" in **Section 6** provides a nuanced view. The comparison in

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(Autoregressive, Diffusion, Parallel) in **Figure 3** ensures that the findings are robust across model architectures and not specific to a single decoding strategy.

Weaknesses

- **Dependence on Hand-Crafted Symbolic Features (Φ):** The central metric of "rule width" heavily relies on the pre-definition of a domain-level feature set Φ (**Section 3.2, Page 2**). As acknowledged in the future work, this limits the immediate applicability of the method to real-world, unstructured environments where such symbolic features are not known *a priori*. The calculated width is a property of the *symbolic abstraction* engineered by the authors, not necessarily the inherent complexity of the raw visual input. If the feature set Φ were defined differently (e.g., fewer combined features vs. more atomic ones), the width—and thus the granularity classification—would change, potentially altering the U-shaped conclusion.
- **Gap Between Symbolic Width and Neural Planning:** While the width metric ($\text{IW}(k)$) rigorously defines difficulty for a breadth-first search algorithm, the paper assumes this translates directly to difficulty for a neural VLA policy. However, neural networks do not perform explicit IW search. **Figure 9** shows that attention to language is generally low. It is possible that the "Medium" difficulty spike (**Figure 3**) is not due to "width 1" being inherently harder for a neural net, but rather because "Medium" instructions (e.g., "soak the rag" without mentioning "pick up" first, as in **Figure 2**) might be pragmatically unnatural or statistically less distinct in the training distribution compared to the explicit steps of Fine or the goal-driven nature of Coarse. The paper does not fully rule out that the U-shape is an artifact of how the specific rules for "Medium" were constructed.
- **Simulation-to-Real Gap:** The experiments are conducted in Mini-BEHAVIOR, a grid-world environment with simplified perception and kinematics. While this is justified for controlled analysis (**Section 4**), the claim that coarse instructions induce language ignoring due to "high visual predictability" (**Section 6**) might be overstated for real-world robotics. In complex real-world scenes, visual predictability is significantly lower than in a grid world with full state observability. The "language ignoring" conclusions might therefore be specific to environments where the next optimal action is easily inferable from vision alone.
- **Lack of Analysis on Instruction Diversity:** The template-based instruction generation (**Section 3.5**) ensures uniformity but eliminates the natural linguistic

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architectures but does not compare against a baseline that explicitly uses the symbolic planner (the Oracle) with noise. Comparing the VLA's failure modes (**Figure 7**, stagnation vs. regression) to a planner with limited horizon or resource constraints would strengthen the claim that the VLA is indeed struggling with *planning width* rather than just memorization failure.

- **Unclear Definition of "Medium" Granularity:** In **Figure 2**, the Medium instruction "If you're holding an unsoaked rag, soak it" ($\$r_{1b}\$$) has width 2. However, the text in **Section 5.1** states "Medium" usually corresponds to $\$w=1\$$. This inconsistency in definition makes it harder to interpret the U-shape. Is the dip at "Medium" because it mixes width 1 and 2, or because it represents a "valley of ambiguity" where the instruction is neither a high-level goal nor a low-level step?

Potentially Missing Related Work

1. **Fang, I., Zhang, J., Tong, S.** (2025): *From Intention to Execution: Probing the Generalization Boundaries of Vision-Language-Action Models*. This work investigates generalization boundaries in VLAs, which is directly relevant to the paper's claims about asymmetric generalization across granularities. It should be discussed in Section 2 or 5.
2. **Hao, Y., Chen, Y., Fan, C.** (2025): *Simulation to Rules: A Dual-VLM Framework for Formal Visual Planning*. This paper connects simulation and formal planning rules using VLMs. Given the submitted paper's focus on "rules" and "planning width," this is a critical related work to position against.
3. **Su, Y., Ling, Z., Shi, H.** (2024): *ActPlan-1K: Benchmarking the Procedural Planning Ability of Visual Language Models in Household Activities*. Since the submitted paper introduces a benchmark (Mini-BEHAVIOR-Gran) for planning/actions, ActPlan-1K is a relevant benchmark comparison that should be acknowledged in the Introduction or Related Work.
4. **Zhang, S., Qiao, Y., Wang, Q.** (2025): *FlexVLN: Flexible Adaptation for Diverse Vision-and-Language Navigation Tasks*. This work addresses adaptability in vision-language tasks, which relates to the "instruction granularity" adaptation discussed in the submission.

Score

8 Strong Accept

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reliance on symbolic features and the simulated grid-world environment, the conceptual novelty and the depth of the analysis (particularly on language ignoring) outweigh these weaknesses. The paper is well-executed and likely to stimulate further research into structured instruction design.

Are the Results in This Paper Easily Reproducible?

Credible

The authors introduce a new benchmark, *Mini-BEHAVIOR-Gran*, and explicitly describe the rule generation process, the feature sets, and the experimental sweep across multiple model architectures. The methodology section (**Section 3**) provides the mathematical definition of rule width, and **Figure 1** outlines the pipeline clearly. Although the code is not linked (likely due to anonymity), the description of the benchmark generation and the use of the established Mini-BEHAVIOR environment make the results reproducible in principle. The inclusion of dataset statistics and training details in the main text and potentially appendix supports credibility.

Questions for the Author Response

- 1. Real-world Feature Extraction:** How sensitive is the "rule width" metric to the choice of the feature set Φ ? If you were to apply this to a real-world dataset (e.g., Open X-Embodiment) where Φ must be learned or inferred, how would you define the "ground truth" width to validate your U-shape findings?
- 2. Medium Granularity Clarification:** In **Figure 2**, the Medium example r_{1b} is labeled with width $w=2$, but **Section 5.1** implies Medium corresponds largely to $w=1$. Could you clarify the exact width distribution of the "Medium" bin? Is the performance drop in the Medium regime driven by the specific mix of widths or by the pragmatic unnaturalness of those instructions?
- 3. Language Ignoring vs. Visual Predictability:** Regarding **Figure 8**, have you analyzed cases where visual predictability is low (e.g., high occlusion or ambiguous states)? Does the "language ignoring" effect diminish for Coarse instructions in such high-uncertainty states?
- 4. Human Evaluation:** Have you considered validating the "naturalness" of the template-generated instructions with human annotators? It would be valuable

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setting, or does the VLA largely ignore context?

Confidence

Note: IJCAI2023 makes confidence level visible to other reviewers and area chairs, but not to authors. I am generally familiar with this topic, but never worked on it myself

Presumed Reviewer Expertise

Note: IJCAI2023 makes reviewer expertise visible to other reviewers and area chairs, but not to authors. This is a posthoc summary of reviewer profile according to the given review above. The reviewer has a strong background in embodied AI, robotic planning, and vision-language models. They are familiar with classical planning concepts (like Iterated Width) and their application to modern neural architectures. They critically evaluate experimental rigor, particularly in benchmark design and variable isolation, and value theoretical formalization of empirical phenomena.

Reviewed on January 20, 2026  Gemini-3-Pro-Preview

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